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RESEARCH

Functional and ultrasonographic evaluation of high intensity laser therapy and Kinesio taping in stroke patients with hemiplegic shoulder pain: a randomized controlled trial

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Abstract

Hemiplegic shoulder pain (HSP) is a common post-stroke complication that limits functional recovery and rehabilitation participation. The objective of this study was to evaluate and compare the clinical, functional, and ultrasonographic superiority of High-Intensity Laser Therapy (HILT) and Kinesio Taping (KT) over each other, as well as in comparison to conventional therapy (COT). The study was designed as a prospective, double-blind, randomized controlled trial. Participants were allocated into three equal groups ($n=11$ each group): Group 1 (Control Group) received COT with sham HILT and sham KT, Group 2 (HILT Group) received COT with active HILT and sham KT, and Group 3 (KT Group) received COT with sham HILT and active KT. COT was performed five days per week for three weeks, HILT was administered in 10 sessions, and KT was applied every other day for three weeks (nine sessions). The primary outcome measure of the study was the Shoulder Pain and Disability Index (SPADI) score. Secondary outcome measures included pain intensity assessed using the Visual Analog Scale (VAS), passive shoulder range of motion (pROM), ultrasonographic findings, Brunnstrom Recovery Stage (BRS), the Functional Independence Measure (FIM), and the Health Assessment Questionnaire (HAQ). All measures were evaluated at baseline (pre-treatment) and immediately following the intervention period (post-treatment). Data were analyzed using repeated measures ANOVA and McNemar tests. Groups were sociodemographically comparable ($p>0.05$). Post-treatment, all groups exhibited significant improvements in shoulder flexion, extension, external rotation, and BRS, VAS, SPADI, FIM, and HAQ scores ($p<0.05$). No significant changes occurred in ultrasonographic findings, shoulder abduction, or internal rotation. Notably, HILT demonstrated superior VAS and SPADI improvements compared to other groups ($p<0.05$). HILT provides rapid, clinically significant pain relief and functional recovery in HSP without adverse effects, offering a safe and superior therapeutic option.

Keywords Hemiplegic shoulder pain · High-intensity laser therapy · Kinesio taping · Stroke rehabilitation · Shoulder ultrasonography · Respiratory function

Introduction

Stroke is a leading cause of global morbidity, with a 1.7% prevalence reported among Turkish adults [1]. A frequent complication is hemiplegic shoulder pain (HSP), affecting 22–47% of patients and negatively impacting functional recovery, psychological well-being, and healthcare costs [2]. The multifactorial etiology of HSP ranging from rotator cuff pathologies and spasticity to central sensitization necessitates a multidisciplinary therapeutic approach [3]. Beyond conventional rehabilitation and pharmacological interventions [4, 5], high-intensity laser therapy (HILT) and

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kinesio taping (KT) have emerged as prominent, non-invasive treatment modalities in current literature.

Photobiomodulation (PBM) is a non-invasive modality that induces analgesia and vasodilation by stimulating peripheral tissues and immune responses [6]. Unlike low-level laser therapy, HILT delivers higher peak power (> 500mW), ensuring faster energy penetration into deep structures [7]. While HILT shows promise for various shoulder pathologies including adhesive capsulitis and HSP, current evidence remains inconclusive due to small sample sizes and a lack of long-term follow-up data [8–10].

Kinesio Taping (KT) utilizes thin, elastic tape to generate cutaneous traction, elevating the epidermis to reduce mechanoreceptor stimulation and enhance vascular and lymphatic flow [11]. While KT aims to modulate pain, improve joint stability, and influence muscle function, its clinical efficacy in HSP remains debated. A meta-analysis by Ghozy et al. [12] reported that KT was not superior to placebo or steroid injections for shoulder pain. Conversely, Deng et al. [13] found KT effective in reducing subluxation and improving functional scores in HSP patients. Due to these conflicting results, no consensus has been reached regarding the optimal role of KT in treatment of HSP.

In the rehabilitation of patients with HSP, HILT offers a non-invasive, easily applicable, and time-efficient option, KT serves as a versatile therapeutic modality utilized for various clinical objectives, such as muscle activation/inhibition and joint stabilization. To the best of our knowledge, there are no existing studies comparing the efficacy of HILT and KT in HSP. The objective of this study was to evaluate and compare the clinical, functional, and ultrasonographic superiority of HILT and KT over each other, as well as in comparison to conventional therapy.

Method

Study design

The study included patients diagnosed with post-stroke hemiplegia who received either inpatient or outpatient treatment at Gaziantep University's Department of Physical Medicine and Rehabilitation between November 2022 and September 2023. Ethical approval for this study was obtained from Gaziantep University's Clinical Research Ethics Committee prior to study initiation (Approval No: 2022/365, Date: 09 November 2022). The study was conducted as a double-blind, randomized controlled trial with a progressive design. All clinical, functional, and ultrasonographic assessments were performed at baseline and post-treatment by an independent physician specialized in physical medicine and rehabilitation, who remained blinded to the treatment allocations. To ensure participant blinding, patients were not informed whether they received active or sham HILT and KT applications. The treatment protocols were administered by a single physical therapist who was aware of the group assignments but was not involved in the assessment process. Written informed consent was obtained from all participants. The research complied with the 1964 Helsinki Declaration and all subsequent amendments, including its ethical principles and guidelines. During the reporting phase, we adhered to the Consolidated Standards of Reporting Trials (CONSORT) guidelines.

Eligibility criteria

The inclusion criteria were as follows: age between 18 and 75 years, a diagnosis of post-stroke hemiplegia by a neurologist based on clinical and radiological findings, hemiplegic shoulder pain on the affected side with a Visual Analog Scale (VAS) score of ≥4, stable medical and neurological status and a Mini-Mental State Examination (MMSE) score of >24 to ensure adequate cognitive function for assessments, interviews, and treatment compliance The exclusion criteria were as follows: neurological instability or diagnosis of neurological disorders (e.g., multiple sclerosis, Parkinson's disease, Alzheimer's disease, dementia), diagnosed rheumatic disease, presence of shoulder arthritis, history of allergic reactions to KT, conditions interfering with treatment application or evaluation (open wounds, rashes, burns, or orthopedic deformities in the affected shoulder region, participation in any rehabilitation program within the last 6 months, botulinum toxin type A (Btx-A) administration for spasticity within the last 6 months, unstable metabolic disorders, and pressure ulcers severe enough to impede rehabilitation. Participants were withdrawn from the study in the event of allergic reactions, non-compliance with the treatment or exercise protocols, metabolic instability, or recurrent stroke during the intervention period.

The eligibility of the participants and all pre- and post-treatment assessments were conducted by an independent physician specialized in physical medicine and rehabilitation.

Of 48 consecutive stroke patients with HSP screened, 12 were excluded: six had received treatment within the past six months, two had prior Btx-A administration, three were non-compliant, and one had pressure ulcers. The remaining 36 patients were equally randomized into three groups via computer-generated block randomization with concealed allocation. Three participants withdrew during the intervention (two due to recurrent stroke and one for KT allergy), leaving 33 patients who completed both baseline and post-treatment evaluations.

Study groups and interventions:

The participants were divided into three groups:

Group 1 (Control Group) Received conventional therapy (COT) plus sham HILT and sham KT.

Group 2 (HILT Group) Received COT plus active HILT and sham KT.

Group 3 (KT Group) Received COT plus sham HILT and active KT.

All groups received a COT program consisting of therapeutic electrical stimulation, neurodevelopmental technique based stretching and passive shoulder range of motion (pROM) exercises, and occupational therapy. The program was administered in 5 sessions per week for 3 consecutive weeks (15 total sessions).

- Stretching exercises: Static stretching protocol (60-second hold, 5 repetitions, 3 sets).
- pROM exercises: 20 repetitions, 3 sets.

Occupational therapy, including gross and fine motor skill training, was administered by the same therapist. Details of the COT protocol are provided in the supplementary file [14].

High intensity laser therapy protocol

In addition to the conventional treatment protocol, HILT was applied over the following days. In the first four sessions, the following parameters were used: 25 Hz, 8 W, 12 J/cm², 300 J/2.5 min in total over a 25 cm² area in analgesic mode, scanning the area circularly from the outside inwards. Subsequently, six sessions were applied in biostimulatory mode, scanning longitudinally with a 3 cm distance between the skin and the probe at 7 W, 100 J/cm², 2,500 J/6 minutes in total over a 25 cm² area. The BTL-6000 high-intensity laser device was used for the HILT protocol. The sham HILT procedure was performed with the laser device inoperative (output power set to 0 J/cm²) to ensure the delivery of zero active radiation [15].

Kinesio taping protocol

In addition to the conventional treatment programme, nine KT sessions were performed every other day for three weeks. The protocol included inhibition using the insertion-origo technique involving the long and short tendons and musculature of the supraspinatus and biceps muscles at 15–35% tension, followed by stabilisation using the mechanical correction technique with the glenohumeral joint (including the acromioclavicular joint) in a neutral position at 50–75% tension. Sham KT was performed without tension [16]. The method of KT application is illustrated in Fig. 1.

All treatment protocols (COT, HILT and KT) were administered by the same physiotherapist.

Outcome measurements

The primary outcome measure of the study was the Shoulder Pain and Disability Index (SPADI) score. Secondary outcome measures included pain intensity assessed using the Visual Analog Scale (VAS), passive shoulder range of motion (pROM), ultrasonographic findings, Brunnstrom Recovery Stage (BRS), the Functional Independence Measure (FIM), and the Health Assessment Questionnaire (HAQ). All measures were evaluated at baseline (pre-treatment) and immediately following the intervention period (post-treatment).

Pain intensity was assessed via VAS, a validated 10-cm scale with high clinical sensitivity and strong correlation with multi-dimensional pain metrics [17]. Motor recovery staging was performed using BRS, focusing on spasticity levels and synergistic movement patterns [18]. Functional independence was quantified through the 18-item FIM, evaluating motor and cognitive domains via a 7-point dependence-independence gradient [19]. Overall functional capacity and disability were measured using HAQ, a comprehensive 20-item index demonstrating excellent internal consistency (Cronbach's alpha = 0.97) [20]. Finally, shoulder-specific impairment was evaluated using SPADI,



Fig. 1 Kinesio Taping Application Protocol for Hemiplegic Shoulder Pain

encompassing pain and disability subscales to capture localized functional status [21].

Ultrasonographic evaluation targeted common HSP pathologies: supraspinatus tendinitis/tendinopathy (SpT), long head of biceps brachii tendinitis/tendinopathy (BiT), and subacromial/subdeltoid bursitis (SASD). Additionally, the acromion-greater tubercle (ACGT) distance was measured in millimeters [22, 23]. SpT was defined by diffuse tendon thickening or hypoechoic/anechoic areas within the fibrillar structure. BiT was identified by decreased echogenicity and a tendon sheath width > 8 mm, while SASD was considered significant if bursal width exceeded 2 mm [23]. All assessments were performed by an experienced physiatrist using a 7–12 MHz linear probe (Logiq P5, GE Medical Systems, WI, USA).

Statistical analysis

Descriptive statistics were presented as mean ± standard deviation (SD) for numerical variables and frequency (percentage) for categorical variables. The normality of data distribution was confirmed using the Shapiro-Wilk test. Data were found to be normally distributed ($p > 0{,}05$). To ensure the study was adequately powered to detect clinically meaningful differences, an a priori power analysis was conducted using G*Power software (version 3.1.9.7). Based on a previously reported large effect size (d = 1.26) for SPADI scores in similar populations (Korkmaz et al., 2022), a minimum of 11 participants per group was required to achieve a statistical power of 0.80 with a significance level of $\alpha = 0.05$. Consequently, a balanced design with an equal number of participants in each group was employed.

To evaluate therapeutic efficacy, a two-way mixed-model repeated measures Analysis of Variance (ANOVA) was performed. The primary focus of this analysis was the Group x Time interaction effect, which assesses whether the rate of clinical improvement over time differed significantly between the groups. This approach was preferred over simple post-treatment between-group comparisons to account for baseline variability and to capture the longitudinal trajectory of recovery. To account for potential violations of sphericity, the Greenhouse-Geisser correction was applied to all F-tests involving the time factor. For each outcome measure, partial eta-squared was reported as an estimate of effect size, providing a standardized measure of the clinical impact. Changes in categorical ultrasonographic findings were analyzed using the McNemar test. All statistical procedures were conducted via SPSS version 22.0, with significance set at $p < 0.05$.

Results

A total of 36 patients were enrolled in the study. No statistically significant differences were observed in sociodemographic characteristics (Table 1). During treatment, two patients were excluded due to new onset stroke and one patient was withdrawn because of KT allergy. The remaining 33 patients completed the intervention, and their pre- and post-treatment outcomes were analyzed. The flowchart detailing the allocation of participants is presented in Fig. 2.

Of the cohort, 3 patients had hemorrhagic stroke, while 33 had ischemic stroke. Hemiplegia was right-sided in 17 patients and left-sided in 16 patients. The group comprised 21 males and 12 females, with a mean age of 59 ± 12.46 years

Table 2 presented the pROM measurements of the study participants, while Table 3 presented the activities of daily living scores and clinical assessment measures, including within-group and between-group differences after treatment.

All treatment groups showed statistically significant improvement in shoulder passive flexion, extension and external rotation ($p < 0.05$), but no significant between-group differences were observed ($p > 0.05$). All groups demonstrated statistically significant improvements in the BRS, VAS, FIM, HAQ and SPADI scales after treatment ($p < 0.001$). In the HILT group, statistically significant differences were observed in VAS and SPADI disability scores compared to the COT and KT groups.

The ultrasonographic evaluation findings of the patients are presented in Tables 4 and 5. No statistically significant changes were observed in SpT, BiT, SASD and (ACGT) measurements before and after treatment in any of the groups ($p > 0.05$).

Table 1 Baseline Sociodemographic Characteristics of Study Participants

Parameters		COT	COT+HILT	COT+KT	*P* value
AGE (years)		58.27 ± 11.62	56.55 ± 12.4	64.09 ± 13.16	0.341
GENDER	Male(n)	7	6	8	0.673
	Female(n)	4	5	3	
BMI		28.78 ± 3.71a	26.31 ± 5.65ab	23.83 ± 3.88b	0.516
HEMIPLEGIC SIDE	Right(n)	3	7	7	0.144
	Left(n)	8	4	4	

No significant differences were found in the sociodemographic characteristics between the groups. Abbreviations, *COT* Conventional treatment, *HILT* high intensity laser therapy, *KT* kinesio taping.

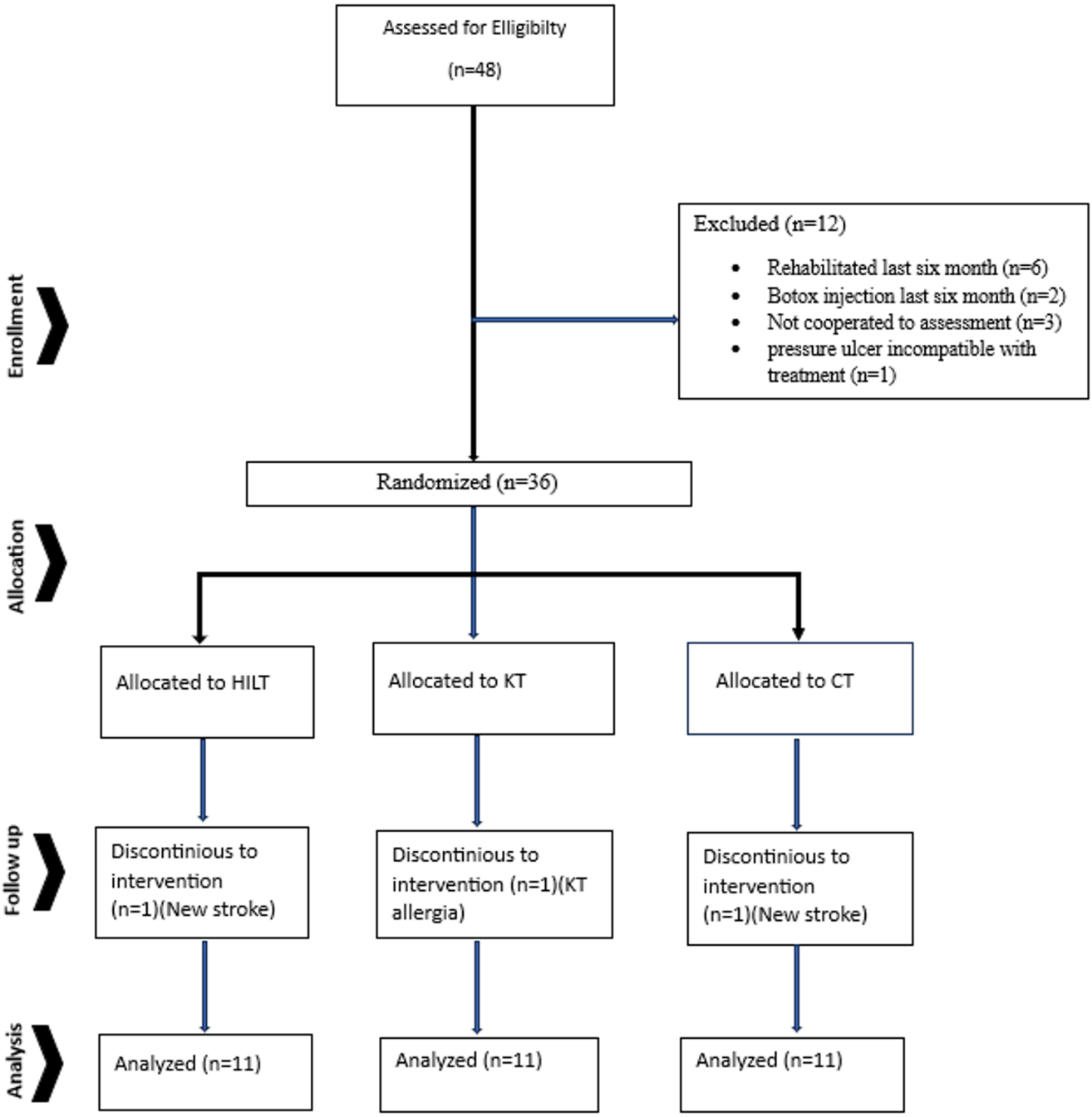


Fig. 2 Participant Flow Diagram

Discussion

HSP negatively impacts rehabilitation success in stroke patients. The use of different treatment options strengthens the physiatrist's hand in management. In current musculoskeletal pain treatment, both HILT and KT are observed as strong options. KT is an easily accessible, cost-effective treatment method that offers application convenience [24, 25]. Compared to LILT, HILT offers advantages such as shorter treatment duration and deeper tissue penetration [8]. However, its clinical utility is limited by disadvantages including higher costs and limited availability across treatment centers [26]. In this study, we aimed to investigate the efficacy of HILT and KT, their superiority over conventional treatment, and to observe the anatomical recovery process.

Systematic reviews have yielded conflicting evidence regarding KT's efficacy, with some trials showing significant pain reduction in musculoskeletal disorders [24] while

Table 2 Presents the comparison of participants pre-treatment and post-treatment shoulder passive range of motion measurements

Parameters	Groups	*N*	Pre-treatment (Mean± SD)	Post-treatment (Mean± SD)	Mean Difference (95% CI)	*p*	η_p^2
Abduction	**COT**	11	101.82±39.26	106.36±34.87	104.09(89.21-118.97)	0.912	0.006
	COT+HILT	11	95.91±13.19	103.64±18.18	99.77(84.89-114.65)		
	COT+KT	11	110.45±19.29	116.36±18.04	113.40(98.53-128.28)		
p			0.055 (η_p^2 ; 0.117)				
Flexion	**COT**	11	115.27±30.16	121.27±33.92	118.27(102.70-133.84)	0.112	0.136
	COT +HILT	11	109.09±25.77	124.55±19.55	116.81(101.24-132.38)		
	COT+KT	11	120.45±25.44	123.64±19.24	122.04(106.47-137.61)		
p			0.002* (η_p^2 ; 0.278)				
Extension	**COT**	11	47.27±25.531	51.64±24.36	49.45(40.63–58.27)	0.296	0.078
	COT+HILT	11	44.55±4.156	46.82±2.52	45.68(36.86–54.50)		
	COT+KT	11	48.18±2.523	48.64±2.33	48.40(39.58–57.22)		
p			0.025* (η_p^2 ; 0.156)				
Internal Rotation	**COT**	11	83.64±11.201	79.55±16.50	81.59(73.04–90.13)	0.052	0.179
	COT +HILT	11	71.64±21.271	82.27±15.71	76.95(68.41–85.49)		
	COT+KT	11	83.36±11.851	82.73±14.72	83.04(74.50-91.58)		
p			0.430 (η_p^2 ; 0.021)				
External Rotation	**COT**	11	63.64±29.757	71.64±17.07	67.63(56.08–79.18)	0.450	0.052
	COT+HILT	11	66.82±23.374	78.00±15.71	72.40(60.85–83.96)		
	COT+KT	11	70.09±21.121	71.36±14.50	70.72(59.17–82.27)		
p			0.043* (η_p^2 ; 0.130)				

A significant improvement was observed in passive shoulder flexion, extension, and external rotation in all groups after the treatment. No significant differences were found between the groups

Values are presented as mean±SD. p: Significance of intra-group change (pre- vs. post-treatment); p: Group × Time interaction based on two-way mixed ANOVA with Greenhouse-Geisser correction; $\eta_p^{2:}$; Partial eta-squared; $p<0.05$ indicates statistical significance

SD Standard deviation, *Cl* Confidence Interval, *COT* Conventional treatment, *HILT* high intensity laser therapy, *KT* kinesio taping

others demonstrate outcomes comparable to placebo [27–29]. For example, Deng et al. reported that KT improved shoulder subluxation, pain control and activities of daily living in patients with HSP [13]. Huang et al. found KT to be more effective than sham taping in improving SPADI scores, shoulder flexion, and internal/external rotation in their study of HSP patients [30]. In another study, Yim et al. found the KT group to be more effective than the sham KT group in HSP treatment for improving subluxation distance, active ROM, VAS, SPADI, and Modified Barthel Index scores [31]. In contrast, Hochsprung et al. compared KT and neuromuscular electrical stimulation (NMES) with COT in their study of HSP patients. After 24 weeks of follow-up, they found no superiority of either KT or NMES over COT [32]. Consequently, no consensus exists in the literature regarding KT's efficacy in HSP treatment. In the present study, although KT intervention yielded significant improvements in passive shoulder flexion, extension, external rotation, and BRS, VAS, FIM, HAQ, and SPADI scores, it demonstrated no clinical superiority over COT or HILT. This outcome may be attributed to our specific KT application focusing on muscle inhibition and joint stabilization. Given that reduced muscle tone and shoulder joint instability are primary etiological factors in HSP, the lack of superiority might also stem from the inherent difficulties in standardizing KT protocols. Factors such as the duration of tape application, variations in tape width, and the high degree of inter-operator variability make KT a challenging modality to standardize. Consequently, we propose that future research should employ diverse taping techniques with longer follow-up periods and more rigorous, comprehensive study designs to account for these methodological variables.

Current literature suggests that HILT is an effective and safe modality for managing shoulder pain in various clinical conditions [33, 34]. In their study on patients diagnosed with subacromial impingement syndrome, Yılmaz et al. demonstrated that HILT was effective in reducing pain, improving functionality and quality of life, and enhancing muscle strength [35]. In their meta-analysis of five randomized controlled trials, Ortiz et al. reported that HILT had a positive effect on pain levels and disability scores in patients with frozen shoulder, though it did not significantly contribute to improvements in shoulder range of motion [36]. In another randomized controlled trial involving patients with HSP, Abdelhakiem et al. significantly greater improvements in pain and functionality scores, as well as handgrip strength, in the HILT group compared to the conventional therapy group post-treatment [15]. In their study of patients with HSP and concomitant partial thickness rotator cuff tears

Table 3 Shows the activities of daily living scores and clinical assessment measures, including within-group and between-group differences following treatment

Parameters	Groups	*N*	Pre-treatment (Mean± SD)	Post-treatment (Mean± SD)	Mean Difference (95% CI)	*p*	η_p^2
BRS	**COT**	11	4.18±1.88	4.36±1.69	4.27(3.24–5.30)	0.287	0.080
	COT +HILT	11	2.55±1.57	3.00±1.54	2.77(1.74–3.80)		
	COT+KT	11	2.64±1.85	3.18±1.60	2.90(1.87–3.94)		
p			<0.001* (η_p^2 ; 0.360)				
VAS	**COT**	11	5.82±1.25	4.82±2.48	5.31(4.18–6.45)	0.037*	0.198
	COT+HILT	11	9.36±3.38	6.00±2.04	7.68(6.54–8.81)		
	COT+KT	11	7.36±1.28	5.64±1.36	6.50(5.36–7.63)		
p			<0.001* (η_p^2 ; 0.510)				
FIM	**COT**	11	80.64±24.38	86.09±21.78	83.36(66.41–100.30)	0.669	0.026
	COT +HILT	11	56.73±32.55	65.00±29.92	60.86(43.91–77.80)		
	COT+KT	11	59.27±27.55	67.45±29.38	63.36(46.41–80.30)		
p			<0.001* (η_p^2 ; 0.458)				
HQA	**COT**	11	1.46±0.64	1.26±0.68	1.36(0.93–1.80)	0.576	0.036
	COT+HILT	11	1.49±0.59	1.28±0.57	1.39(0.95–1.82)		
	COT+KT	11	1.59±0.86	1.29±0.88	1.44(1.01–1.88)		
p			<0,001* (η_p^2 ; 0,540)				
SPADI Pain	**COT**	11	44.18±27.23	32.727±24.53	38.45(24.42–52.48)	0.457	0.051
	COT +HILT	11	70.09±24.20	48.236±29.92	59.16(45.13–73.19)		
	COT+KT	11	60.73±24.52	44.45±20.37	52.59(38.55–66.62)		
p			<0.001* (η_p^2 ; 0.448)				
SPADI Disability	**COT**	11	37.59±28.75	32.74±21.33	35.17(20.76–49.57)	0.030*	0.208
	COT +HILT	11	60.70±17.84	35.61±29.92	48.16(33.75–62.56)		
	COT+KT	11	46.16±26.56	44.45±28.45	38.78(24.37–53.18)		
p			<0.001* (η_p^2 ; 0.461)				

A significant improvement was observed in daily living scores and clinical assessment measures across the groups post-treatment. In the HILT group, a statistically significantly greater improvement was noted in the VAS and SPADI disability scores compared to the KT and COT groups

Values are presented as mean±SD. p: Significance of intra-group change (pre- vs. post-treatment); p: Group × Time interaction based on two-way mixed ANOVA with Greenhouse-Geisser correction; $\eta_p^{2:}$; Partial eta-squared; $p<0.05$ indicates statistical significance

SD Standard deviation, *Cl* Confidence Interval, *COT* Conventional treatment, *HIlT* high intensity laser therapy, *KT* kinesio taping, *BRS* Brunnstrom Recovery Scale, *VAS* Visual Analog Scale, *FIM* Functional Independent Measure, *HAQ* Health Assessment Questionnaire, *SPADI* Shoulder Pain and Disability Index

Table 4 Pre-treatment and post-treatment evaluation results of SpT, BiT, and SASD in study participants

Participants (*n*)	Pre-treatment	Post-treatment	*P* value
With-SpT	32	27	0.125
Without-SpT	1	6	
With-BiT	23	16	0.065
Without-BiT	10	17	
With-SASD	6	4	0.625
Without-SASD	27	29	

No significant differences were found in the SpT , BiT and SASD findings among the groups after the treatment

SpT Supraspinatus tendinitis/tendinopathy, *BiT* Biceps brachii long head tendinitis/tendinopathy, *SASD* Subacromial/subdeltoid bursitis

(PTRCT), Korkmaz et al. observed statistically significant post treatment improvements in the HILT group compared to controls across multiple outcome measures: VAS pain scores, FIM, BRS, SPADI, Nottingham Health Profile quality of life assessment and ultrasonographic measurements of PTRCT dimensions [10]. In our study, the HILT therapy group demonstrated post-treatment improvements in passive shoulder flexion, extension, and external rotation range of motion, along with enhanced scores on the BRS, VAS, FIM, HAQ, and SPADI. Notably, we observed statistically significant greater improvements in VAS and SPADI disability scores compared to both KT and COT groups. However, no significant between-group differences were found in SPADI pain scores. This observation may be explained by our study's methodology of assessing VAS scores at rest, whereas the significant improvement in SPADI disability scores likely reflects both rapid pain modulation and subsequent functional gains following HILT intervention. Although the HILT group demonstrated significantly greater pain reduction compared to control groups, this clinical improvement was not correlated with observable changes in tissue-level ultrasonographic findings. We suggest that this discrepancy may be attributed to the molecular-level cellular changes, improved tissue perfusion, and modulation of pain mediators induced by PBM, which are often beyond

Table 5 Comparative analysis of ACGT measurements before and after treatment in study participants

Parameters	Groups	Pre-treatment (Mean± SD)	Post-treatment (Mean± SD)	Mean Difference (95% CI)	*p*	η_p^2
ACGT(mm)	**COT**	2.81 ± 0.54	2.74 ± 0.55	2.77(2.45–3.09)	0.455	0.051
	COT+HILT	3.13 ± 0.66	2.92 ± 0.52	3.03(2.70–3.35)		
	COT + KT	2.93 ± 0.61	2.88 ± 0.59	2.93(2.61–3.25)		
p		0,641 (η_p^2 ; 0,007)				

Post-treatment assessment revealed no significant differences in Acromioclavicular Joint Degeneration (ACGT) measurements among the groups

Values are presented as mean ± SD. p: Significance of intra-group change (pre- vs. post-treatment); p: Group × Time interaction based on two-way mixed ANOVA with Greenhouse-Geisser correction; $\eta_p^{2:}$; Partial eta-squared; $p < 0.05$ indicates statistical significance

SD Standard deviation, *Cl* Confidence Interval, *COT* Conventional treatment, *HILT* high intensity laser therapy, *KT* kinesio taping, *ACGT* Acromion-great tubercule measure

the resolution of conventional ultrasonography. Furthermore, the absence of significant structural changes may be linked to the short follow-up duration, which may have been insufficient to capture detectable anatomical remodeling, despite the observed clinical and functional improvements [37, 38]. Our findings suggest that HILT is an effective and safe therapeutic option for HSP management, demonstrating rapid pain relief, enhanced functional outcomes, and no observed adverse effects throughout treatment.

Conclusion

In conclusion, our findings demonstrate that HILT is superior to KT and control groups in reducing pain intensity and improving functional disability, as measured by VAS and SPADI scores. While KT showed no significant therapeutic advantage over other groups, it was associated with an allergic reaction in one participant, whereas HILT demonstrated an excellent safety profile with no adverse effects.

From a clinical perspective, HILT offers a promising, non-invasive, and time-efficient intervention for rapid pain relief in hemiplegic shoulder pain, particularly when a standardized taping protocol for KT remains elusive. To our knowledge, this is the first randomized controlled trial to directly compare these two modalities in this population. Although these results are encouraging, further multicenter studies with larger cohorts and extended follow-up periods are warranted to confirm the long-term durability of these functional gains and structural outcomes.

Limitations

Study limitations include a small sample size and short follow-up, which potentially hindered the detection of long-term functional gains or anatomical remodeling via ultrasonography. The absence of a pure placebo group, lack of patient stratification by post-stroke phase, and the etiological heterogeneity of HSP may have influenced outcomes. Additionally, the potential for expectation bias and the lack of a standardized KT protocol limit the generalization of our findings.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Compliance with ethical standards The study was approved by the Gaziantep University Clinical Research Ethics Committee (No: 2022/365, Date: 09/11/2022).

Consent to participate Written informed consent was obtained from all participants.

Competing interests The authors declare no competing interests.

Clinical trial number Not applicable.

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